Real-Estate Price Prediction

From-scratch decision trees and Pegasos SVMs on bina.az listings

MLE-AI-201 Final Project
AI Academy — Cohort II

18 October 2026

Two tasks, one reproducible pipeline: estimate listing price and classify listings as standard or premium.

Problem and dataset

Task A — regression

Predict listing price in AZN from property, location, and listing features. Report RMSE, MAE, and R^2.

Task B — classification

Predict a binary price tier: premium when price exceeds the training median; standard otherwise. Report precision, recall, F1, and ROC-AUC.

Dataset snapshot

Raw listings	100,775
Clean listings	100,705
Encoded features	178
Split	70 / 15 / 15%
Tier threshold	218,000 AZN

Seeded, proportional sampling within price-rank strata; the tier cutoff is computed from training data only.

Data preparation and leakage controls

Cleaning and feature work

- Enforce AZN currency; remove missing/non-positive prices and apply fixed eligibility ranges.
- Parse building and land area; convert land units and split floor into current and total floors.
- Impute numeric missing values from training data; encode categorical features with training-fitted preprocessing.
- Keep location coordinates, rooms, area, listing flags, and selected categorical fields.

Leakage prevention

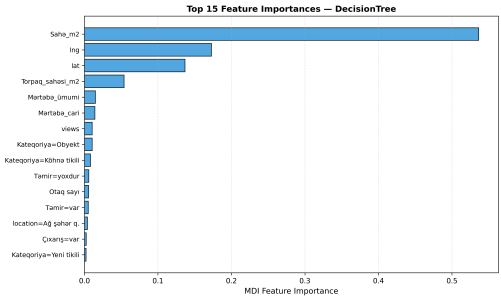
`unit_price` and `total_price` are excluded because they reveal the target. Identifiers, URLs, owner names, addresses, and free text are outside the selected feature set.

Split first; fit imputers, encoders, and scalers on training data only. Validation selects settings; test data is held for final evaluation.

From-scratch decision trees

- Numeric threshold splits chosen by weighted impurity reduction.
- Gini and entropy for classification; variance/MSE for regression.
- Stopping controls include maximum depth, minimum split/leaf counts, and minimum impurity decrease.
- Select tree settings using validation performance only (limited tuning due to $O(\text{candidates}^p)$ complexity).
- Fitting data takes $O(\text{depth} \cdot \text{features} \cdot \text{rows} \cdot \log(\text{rows}))$ time.

Selected scratch settings: regression, MSE, depth 15, leaf 1; classification, entropy, depth 15, leaf 1.



Impurity-based feature importances from the selected tree.

Pegasos SVM and nonlinear features

The scratch SVM minimizes the regularized hinge-loss objective:

$$J(\mathrm{w}, b) = \frac{\lambda}{2}\|\mathrm{w}\|^2 + \frac{1}{n}\sum_{i=1}^{n} \max\bigl(0, 1 - y_i(\mathrm{w}^{\mathsf{T}}\mathrm{x}_i + b)\bigr) .$$

- Pegasos stochastic sub-gradient updates use $\eta_t = 1/(\lambda t)$.
- Standardization is fit on training data; objective and margin diagnostics are available during fitting.
- Random Fourier Features map inputs to an approximate RBF space.

Selected by validation

Linear Pegasos
$\lambda = 0.001$, 500,000 updates

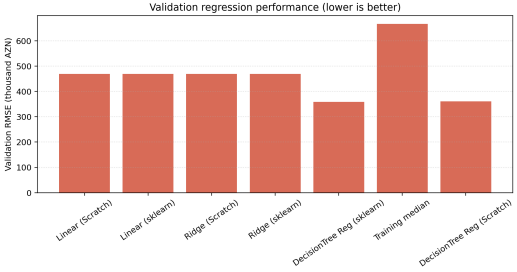
RFF Pegasos
$\gamma = 0.001$, 500 features, 200,000 updates

Evaluation protocol

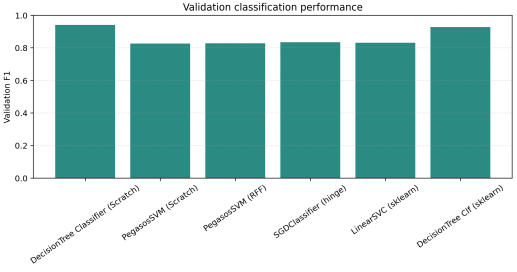
- Fixed seed: 42. Train on 70,493 listings; validate on 15,106; evaluate once on 15,106 held-out test listings.
- Regression: scratch Linear/Ridge, scratch tree, sklearn Linear/Ridge, sklearn tree, and training-median reference.
- Classification: scratch tree and Pegasos (linear/RFF), plus sklearn SGD hinge, LinearSVC, and decision-tree baselines.
- Configuration selection uses training and validation data only; validation metrics and final held-out test metrics are shown separately.

Validation results support model selection; test results provide the final held-out evaluation. Regression errors are in AZN.

Validation plots

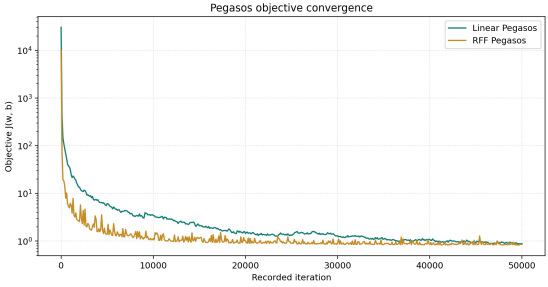


Regression validation RMSE; lower is better.

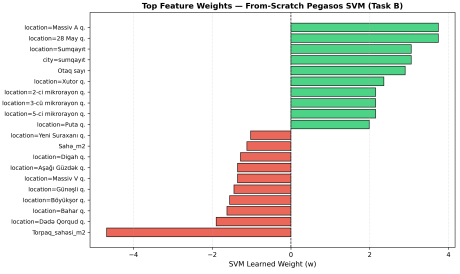


Classification validation F1; higher is better.

SVM diagnostics



The stochastic objective decreases overall; local increases are expected.



Largest positive and negative linear Pegasos weights.

Validation results — regression

Model	RMSE (AZN)	MAE (AZN)	R^2
Linear (scratch)	469,459	138,452	0.489
Linear (sklearn)	469,459	138,452	0.489
Ridge (scratch)	469,459	138,452	0.489
Ridge (sklearn)	469,459	138,452	0.489
Decision tree (sklearn)	**359,068**	86,810	**0.701**
Decision tree (scratch)	360,381	**71,074**	0.699
Training-median reference	666,155	191,467	-0.030

The sklearn tree has the best validation RMSE and R^2; the scratch tree has the lowest validation MAE. Linear and Ridge remain nearly identical.

Validation results — price tier

Model	F1	ROC-AUC
Pegasos (scratch, linear)	0.827	0.907
Pegasos (scratch, RFF)	0.828	0.906
SGDClassifier (hinge)	0.836	0.910
LinearSVC	0.832	0.918
Decision tree (scratch)	**0.942**	0.962
Decision tree (sklearn)	0.928	**0.967**

The scratch tree has the best validation F1, while sklearn's tree has the best ROC-AUC. These validation scores are not the final test estimates.

Test results — regression

Model	RMSE (AZN)	MAE (AZN)	R^2
Linear (scratch)	505,494	139,441	0.450
Linear (sklearn)	505,494	139,441	0.450
Ridge (scratch)	505,495	139,441	0.450
Ridge (sklearn)	505,495	139,441	0.450
Decision tree (sklearn)	401,341	88,675	0.653
Decision tree (scratch)	**380,596**	**71,856**	**0.688**
Training-median reference	691,594	196,612	-0.030

The scratch tree has the lowest test RMSE and MAE among the reported regression models. Linear and Ridge agree closely with their sklearn counterparts.

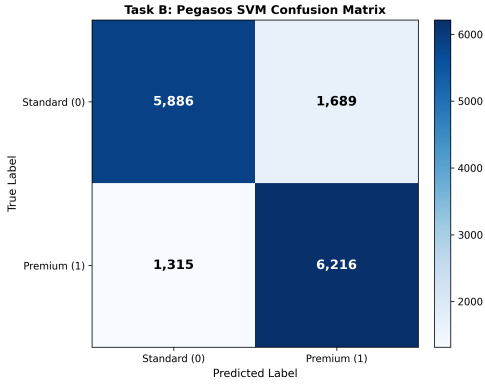
Test results — price tier

Model	F1	ROC-AUC
Pegasos (scratch, linear)	0.826	0.911
Pegasos (scratch, RFF)	0.832	0.908
SGDClassifier (hinge)	0.838	0.910
LinearSVC	0.835	0.920
Decision tree (scratch)	0.941	0.962
Decision tree (sklearn)	**0.930**	**0.967**

The scratch tree leads the listed classifiers in test F1; sklearn's tree has the highest ROC-AUC. Pegasos and linear baselines are competitive but trail the trees on this split.

Error patterns and model selection





Price errors are asymmetric with several large residuals; tier classification is substantially stronger than a trivial baseline, but still misclassifies both classes.

Four-member responsibility split

Mahammadali Babayev— Data and Task A

- Cleaning, preprocessing, feature engineering, and leakage controls.
- Scratch Linear/Ridge via normal equations; sklearn regression models and Task A evaluation.

Ilham Hajiyev — Decision trees

- Scratch Gini/entropy/MSE tree, threshold search, and stopping rules.
- Tree unit tests; Tree HP tuning.

Mahammad Javadzade — SVM

- Scratch Pegasos, objective/margin/support-vector diagnostics.
- Nonlinear RFF or kernel extension and feature scaling.

Fariz Isgandarli — EDA and integration

- EDA and optional geospatial clustering/PCA.
- sklearn classification baselines, metrics, feature importance, error analysis, runner, and repository setup.

Limitations and next steps

Limitations

- One geographic marketplace and one fixed data snapshot.
- A median split discards price magnitude and depends on the training sample.
- Large price residuals remain; aggregate metrics hide local and category-specific error differences.
- Single held-out split; reported values are not fold-averaged.
- Limited tuning due to time and reproducibility constraints.

Next steps

- Validate across time or repeated splits.
- Analyze errors by region, property type, and price range.
- Compare calibrated thresholds and probability quality.
- Document data provenance and monitor for marketplace drift.

Questions?